Anti-theft and Tracking Mechanism for Vehicles using GSM and GPS

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ABSTRACT: There is a drastic increase in the number of crimes involving vehicle theft. This can be avoided by using anti-theft and tracking mechanism using GSM and GPS. In the current scenario it takes minimum two days time to find the lost or the stolen vehicle with the help of police. This mechanism alerts the owner or the user of the vehicle with the message at the time of theft occurred .Moreover this mechanism helps to find the location of the theft vehicle by using GSM and GPS at anytime and also this mechanism stores the message by the application in the mobile. This mechanism helps to prevent and reduce the theft of vehicles. Further these types of devices are less expensive and also affordable for all the vehicle owners.

INTRODUCTION

A colossal increase in the number of vehicles across the world further leads to the increase in the theft of vehicles. Though we have many security mechanisms for vehicles, the crime has been increasing tremendously. At present the vehicles have been provided with the system which alerts the user with the alarm at the time of theft, but the nominal users are still facing the crime. In order to avoid this crime the anti-theft and tracking mechanism for vehicles can be used, which tracks the current location of the vehicle using GPS and the GSM is used for communication. This mechanism intimates the user with an alert message at the time of theft and the user gets the timely updates with the current location of the vehicle. The anti-theft and tracking mechanism for vehicles using GPS is cost effective.

RELATED WORKS

Abubakr S. Eltayeb, Halla O. Almubarak, Tahani Abdalla Attia[1]has discussed that in order to reach their destination ambulances, police force vehicles and fire fighting vehicles are required. These mechanisms are required in order to avoid unnecessary delay. The design and implementation of an automatic pre-emption traffic control system, based on the global positioning system satellites mechanism is used in this paper. The proposed system has a better solution for the delay time experienced on the emergency vehicles along their path. They need infrared transmissions or visual light transmissions to provide the path clearance in advance for it. Traffic light pre-emption systems have not solved completely in this paper it has some issues.

Mohamed A.Abousalem and Edward J. Krakiwsky [2] has involved the engine for processing kinematic data collected from positioning sensors and widely used for multi sensor navigation system and also for the performance of sensor for constituting an integrated system. All these are used with the theory of federated architecture. They use a type of decentralized filters for vehicle navigation using GPS the main goal is required accurate and reliable systems with fault detection and isolation and recovery capabilities.

Anirban Chowdhury, Jhulan Kumar, Abhra Majumder[3], have showed nowadays road traffic and reckless driving is one of the major problem faced by the people. For this problem they provide solution using an passive RFID tags which is electronically control the road vehicles with a limited speed for each type of road traffic. They have used this type of electronic device as it is cost effective because they do not require any external power but it can work with the help of internal powers. The future modification of upgrading in the model can be done. According to the idea they have implemented

the concept, but still there some issues that they have faced because of the passive RFID tags give only economic solutions.

Arghavan Amini1, Reza Monir Vaghefi, Jesus M. de la Garza1, and R. Michael Buehrer[4] has discussed that in order to highlight the performance and quality of transportation network, a method of Intelligence Transportation System(ITS).ITS is needed to provide a accurate location and information about the vehicle. It also provides the integrated localization algorithm which required data from different resources including GPS, radio-frequency identification, vehicle-to-vehicle etc .This localization algorithm is very useful only if all the resources are available. This proposed algorithm will improve the accuracy even if the GPS is weak. The ITS involves the major algorithm called the Integrated and Smart positioning algorithms. The main idea is to calculate the accurate location of the vehicles that are obtained from the RFID tags and GPS satellites.

P.Bagavathy,R.Dhaya&T.Devakumar[5], have focused on GPS(Global Positioning System) based tracking system for vehicles. This system provides the method to the user to track their vehicle remotely through the mobile network. Global Positioning System (GPS), Global System for Mobile communication (GSM) modem and microcontroller are provided with the aim of the users to locate their vehicles with ease and in a convenient manner. The exact position of the vehicle will be displayed using a GUI application. Vehicle Tracking System was used in many ways to track and display the location time in real time.

To overcome the above issues, anti-theft and tracking mechanism can be implemented to .avoid the theft of vehicles by sending the alert message to the user at the time of theft.

ANTI-THEFT AND TRACKING MECHANISM

As usage of vehicles is growing day by day, the theft of vehicles is also growing highly. Our mechanism helps to detect the theft and also used to prevent the vehicle from the theft. The anti-theft and tracking mechanism for vehicles uses GPS system to track the current location of the vehicle which further helps us to prevent the vehicle from theft. Our mechanism consists of the following components namely,

- RFID Reader
- RFID Tag
- Microcontroller
- Vibration Sensor
- GPS
- GSM
- LCD Display

An RFID reader is a network connected device that sends power as well as data and commands to the tags. It acts like an access point for RFID tagged items so that the tags data can be made available. A passive tag is an RFID tag that does not contain a battery; the power is supplied by the reader. The RFID Reader and the RFID Tag is used for the detecting purpose. The RFID Reader is placed in the vehicle and the RFID Tag is placed in the key. When both the RFID Reader and the RFID Tag gets matched with the unique code placed in the reader and the tag, the vehicle gets started. In case, if both the RFID Reader and the RFID Tag does not get matched, the alert message about the theft will be sent to the user mobile phone using GSM. The microcontroller consists of RFID reader, tag, GPS, GSM and the vibration sensor. The vibration sensor is placed in the vehicle and using that the vibration will get sensed when the RFID reader and tag does not get matched. For every five minutes the location will be tracked and updated automatically. So that the vehicle user can easily know about the direction of the vehicle which helps the vehicle user to find their lost vehicle.

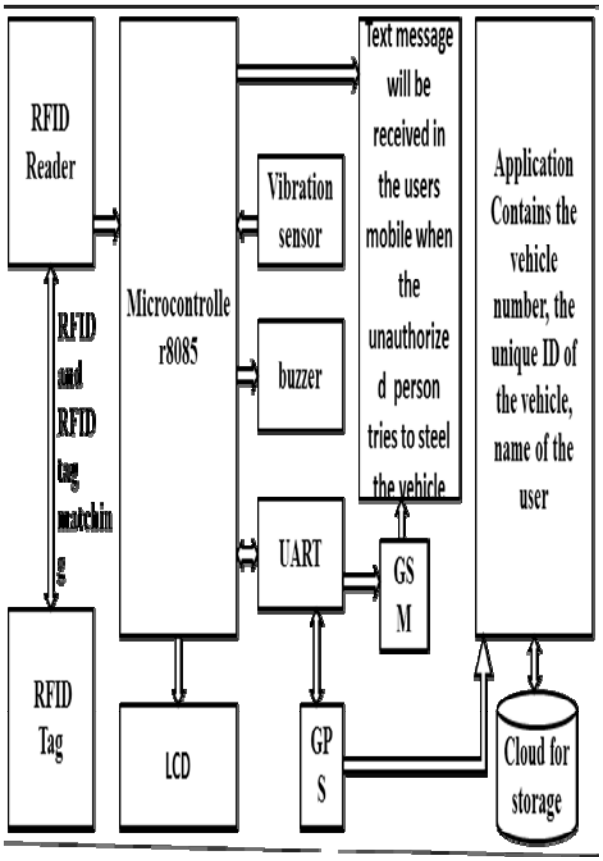

Fig 1: Architecture diagram

Fig 2: GSM module and LCD display

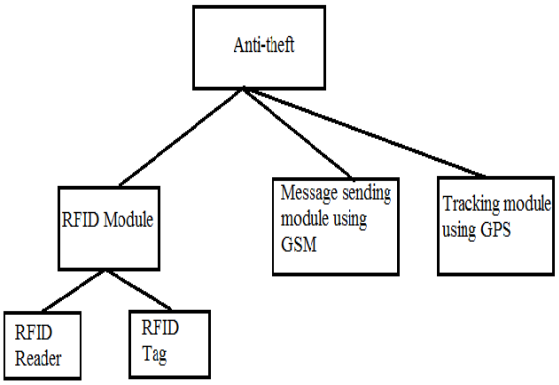

Fig 3 : Logical view of anti-theft mechanism

The anti-theft and the tracking mechanism consist of the process of RFID reader and RFID tag matching. Once the unique id in the tag and the reader matches the alert message will not be sent to the user and if the tag does not get matched, the alert message regarding the theft of the vehicle will be sent to the user mobile using the GSM. This process further continues with the intimation to the user about the current location of the vehicle once every five minutes. This helps the user to easily find their vehicle. Anti-theft mechanism is time efficient and can be used for avoiding the crime of vehicle theft. The GPS system is also used for tracking purpose which helps us to detect the theft. GSM is used for communicating with the user.

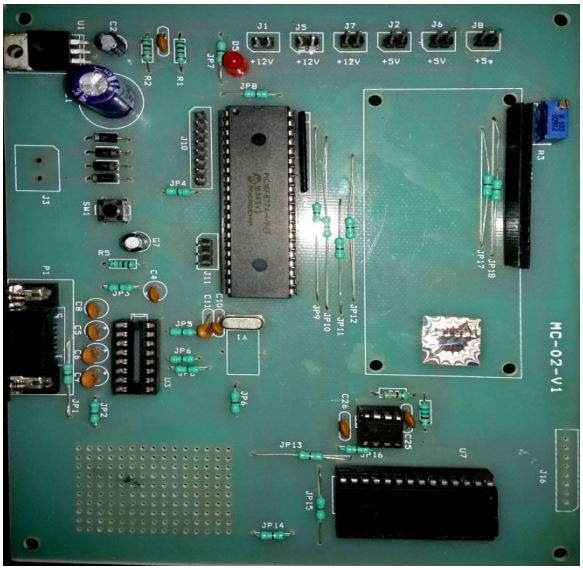

Fig 4:ATMEL 8052 microcontroller board

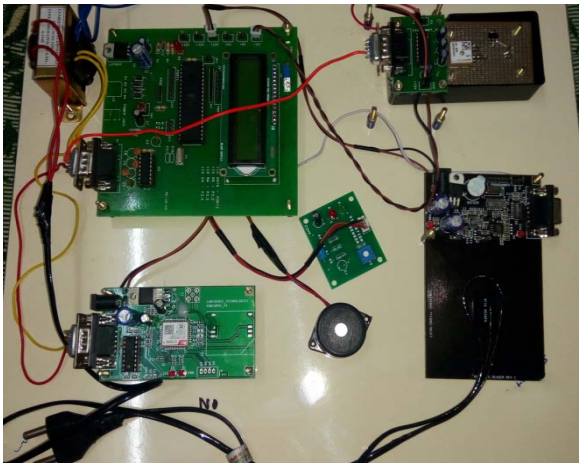

Fig 5: Hardware setup of the kit

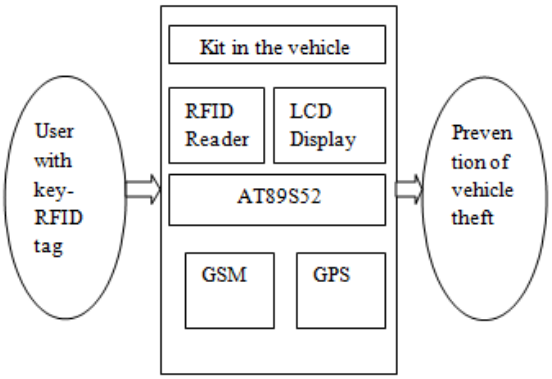

Fig 6:Setup of hardware kit

V REFERENCES

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